

Mahdi Mohammadigohari

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Research Summary

I develop theoretical foundations for deep learning from a function-space and statistical perspective. My work focuses on understanding generalization in deep models beyond parameter-based analyses, using kernel methods, operator-theoretic techniques, and hierarchical constructions of hypothesis spaces. A central theme of my research is to characterize how compositional depth influences regularity and statistical complexity. I develop frameworks that yield dimension-free and depth-uniform generalization guarantees, providing a principled explanation for how deep architectures can remain both expressive and statistically stable. More broadly, my goal is to connect functional analysis and statistical learning theory with modern deep learning, and to build mathematically grounded models for complex learning systems.

Research Interests

Machine Learning Theory; Functional Analysis.

Education

- **Ph.D. in Computer Science (Machine Learning Theory)** Nov 2022 – July 2026
Free University of Bozen–Bolzano, Italy
(Defense scheduled July 2026)
Thesis: *Mathematics of Deep Learning: Generalization in Single-Task and Multi-Task Architectures via Operator-Theoretic and Kernel-Based Methods*
Advisor: Prof. Giuseppe Di Fatta
- **M.S. in Data Analytics** Sep 2020 – May 2022
Rochester Institute of Technology (RIT Dubai)
GPA: 3.7/4
Thesis: *Energy Consumption Forecasting Using Machine Learning* ([online link](#))
Completed all coursework and thesis requirements. Graduation could not be formally finalized due to a visa interruption (documentation available upon request).
Advisor: Prof. Sanjay Modak
- **Ph.D. in Mathematics (Functional Analysis)** Sep 2014 – Mar 2018
Islamic Azad University of Mashhad, Iran
GPA: 18.7/20
Thesis: *Operator Inequalities and Operators Preserving Orthogonality*
Advisor: Prof. Maryam Amyari
- **M.S. in Pure Mathematics** Sep 2003 – Feb 2006
Islamic Azad University of Mashhad, Iran
- **B.S. in Applied Mathematics** Sep 1999 – Sep 2003
Islamic Azad University of Mashhad, Iran

Research Experience

- **Doctoral Researcher in Machine Learning Theory**, Free University of Bozen–Bolzano
2022 – Present
 - *Brownian Kernel Ladders (BKL)*: Introduced a hierarchical function-space framework based on integral RKHS constructions driven by Brownian kernels. Developed an intrinsic complexity functional independent of parameterization and established depth-uniform, dimension-free Gaussian complexity and excess-risk bounds of order $\mathcal{O}(n^{-1/2})$. This shows that increasing depth does not necessarily increase statistical complexity.
 - *Operator-Theoretic Generalization for Multi-Task Deep Learning*: Developed a Koopman operator-based framework for analyzing generalization in deep multi-output models. Introduced Brownian integral RKHSs as an alternative to Sobolev spaces, yielding improved spectral scaling and a unified treatment of injective and non-injective architectures.

Applied Data Science Experience

- **Time Series Modeling and Forecasting**: Developed machine learning models for energy consumption forecasting using real-world data, including statistical and learning-based approaches for temporal prediction.

Teaching Experience

- **Faculty Member / Instructor**, Islamic Azad University (Iran) 2006 – 2020
Courses: Linear Algebra, Real Analysis, Probability Theory, Statistical Inference, Differential Equations, Numerical Analysis.

Research Grants & Funded Projects

- **Post-Lauream Research Grant**, Free University of Bozen–Bolzano, Italy Dec 2025 – Present
Project: *AI Engineering Suite to Support Agile and Efficient Software Engineering (AI4SWEng)*
Principal Investigator: Prof. Giuseppe Di Fatta
Funding: University Research Grant (CUP: I53C250000000006)
Role: Doctoral researcher contributing to theoretical aspects of machine learning and generalization in AI-driven systems.
- **Learning Fair Latent Representations with Multi-Task Deep Learning** Nov 2025 – Dec 2025
Short-term funded research project under the MTDL Start-Up Fund
Focus: Fairness-aware multi-task deep learning and theoretical generalization guarantees for representation learning.

Selected Publications

Machine Learning Theory

- M. Mohammadigohari, G. Di Fatta, G. Nicosia, P. M. Pardalos. *Brownian Kernel Ladders*. Submitted to the Journal of Machine Learning Research (JMLR), 2026.
- M. Mohammadigohari, T. Borsani, G. Di Fatta, G. Nicosia. *Fair Latent Representation Learning by Multi-Task Deep Learning*. Manuscript under review.
- M. Mohammadigohari, T. Borsani, G. Di Fatta. *Operator-Based Generalization Bounds for Multitask Deep Learning*. Manuscript under review.
- M. Mohammadigohari, G. Di Fatta, G. Nicosia, P. M. Pardalos. *On the Koopman-Based Generalization Bounds for Multi-Task Deep Learning*. In: *Machine Learning, Optimization, and Data Science (LOD 2025)*, Lecture Notes in Computer Science, vol. 16468, Springer, 2026. https://doi.org/10.1007/978-3-032-21480-5_25
- M. Mohammadigohari, G. Di Fatta, G. Nicosia, P. M. Pardalos. *Operator-Based Generalization Bound for Deep Learning: Insights on Multi-Task Learning*. In: *Machine Learning, Optimization, and Data Science (LOD 2025)*, Lecture Notes in Computer Science, vol. 16468, Springer, 2026. https://doi.org/10.1007/978-3-032-21480-5_9

Mathematics / Functional Analysis

- M. Mohammadigohari, M. Amyari. *Operator-Valued Parallelism and Norm-Parallelism in Matrices*. *Indian Journal of Pure and Applied Mathematics*, 51(4):1305–1316, 2020.
- M. Mohammadigohari, M. Amyari. *Parallelism in Hilbert $K(\mathcal{H})$ -Modules*. *Tbilisi Mathematical Journal*, 12(2):67–75, 2019.
- M. Mohammadigohari, M. Amyari. *Refinements of the Heinz Inequalities for Operators and Matrices*. *Mathematica Slovaca*, 68(6):1431–1438, 2018.

Technical Skills

- Mathematics: Functional Analysis, Operator Theory
- Machine Learning: Machine Learning Theory, Statistical Learning Theory, Deep Learning
- Programming: Python, R, MATLAB
- Tools: $\text{\LaTeX}$, Git, TensorFlow, scikit-learn
- Languages: English (Proficient), Persian (Native)